Always formalize your pizza before ordering

Ferdinand Equilbey

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Abstract

We describe algebraically the crust of a pizza in arbitrary dimension. Proper volume computations and geometrical reasoning allow to approach how many bites the average client would make to finish it. We believe that in addition to a sauce model, a proper n-dim pizza will be understood in future works.

The toric crust framework

We assume that the crust of the pizza is a *gluing* of two components: a torus (sometimes cheese-filled) and a disk, which supports the garniture: we refer to this as the *toric crust model*, in arbitrary dimension n, writing $\mathscr{C}_{n,k} = \mathbb{T}^n \times \mathbb{D}^k$, where k is the dimension of the disk. Note that, there are $n-1$ possible choices for k, but it seems that $k = n-1$ best reflects our known $3d$ model.[1] We also remark that a $n-1$ torus would be acceptable.

Let $\mathscr{C}$ be the Cartesian product of an n-dimensional torus $\mathbb{T}^n$ and a k-dimensional disk $\mathbb{D}^k$. We define the components as follows. Give a collection of radii $\{r_i\}_{i=1}^n$, the n**-torus** is defined as the product of n one-dimensional circles, written $S^1(r_i)$. We have $\mathbb{T}^n = \prod_{i=1}^n S^1(r_i)$. Secondly, the k**-disk** (or k-ball) of radius R in $\mathbb{R}^k$ is defined by: $\mathbb{D}^k(R) = \{x \in \mathbb{R}^k : \|x\| \leq R\}$.

From this we can rapidly compute the total volume of the crust: the standard relations

$$\mathrm{Vol}(\mathbb{T}^n) = (2\pi)^n \prod_{i=1}^n r_i, \quad \mathrm{Vol}(\mathbb{D}^k) = \frac{\pi^{k/2}}{\Gamma(\frac{k}{2}+1)} R^k$$

can be assembled ine one product, by virtue of the Fubini's theorem. The total volume of crust in dimension $N = n + k$ is then:

$$\mathrm{Vol}(\mathscr{C}) = \frac{2^n \pi^{n+\frac{k}{2}}}{\Gamma(\frac{k}{2}+1)} R^k \prod_{i=1}^n r_i$$

[1] To our knowledge, evidence of other model has been shown.

In the isotropic case where all torus radii are equal ($r_i = r$), the expression simplifies to:

$$\mathrm{Vol}(\mathscr{C}) = \frac{2^n \pi^{\frac{2n+k}{2}}}{\Gamma(\frac{k}{2}+1)} r^n R^k$$

We plot this sequence for growing dimension N in Figure 1. Note in particular that r and R have different impact.

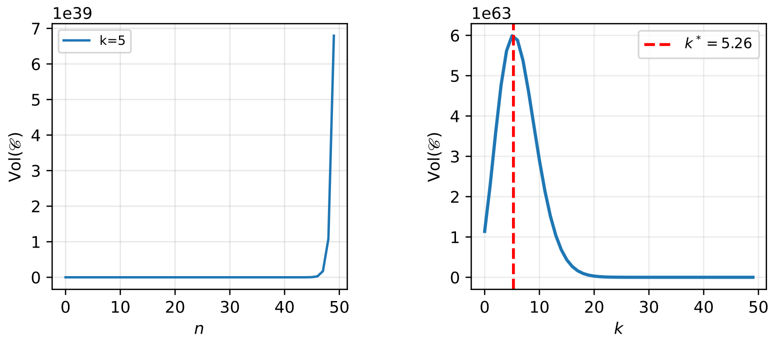

Figure 1: Total volume of crust with respect to growing torus and support disk dimensions.

1 Discussion

In this note, we defined the toric crust and developped a generalized model that will allow to view pizza in an algebraic sense. Although the interest is limited, future work could include computing the best dimension to order, in terms of sauce- or garniture-crust ratios. Applying the same type of reasoning to tasty crousties, shawarmas, burgers, and other fast-food like dishes, that we sadly posit to be the most prominent in Milky-way alike galaxies, could be beneficial for future consumers.